

Muhammad Zain

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EDUCATION

University of Calgary

Bachelor of Science in Software Engineering

Calgary, AB

September 2022 - April 2027

- **Relevant Coursework:** Web and Database Systems, Machine Learning, Algorithms and Data Structures, Software Design Principles and Testing, Object-Oriented Programming, Operating Systems, Networks and Embedded Systems

SKILLS

- **Programming:** JavaScript, Python, Java, C++, SQL, HTML/CSS, RISC-V Assembly, LaTeX
- **Frontend and Backend:** React (Hooks), Node.js, Next.js, Flask, REST APIs, Responsive UI, Tailwind CSS
- **Databases:** Postgres (PostgreSQL), MySQL, SQLite, Query Optimization, Indexing, Database Design
- **Tools and Platforms:** Git/GitHub, Vercel, CMake, Qt (C++), Java Swing, Arduino

EXPERIENCE

Enactus UCalgary - Paragon: Wildfire Project

Software Development Delegate

Calgary, AB

September 2025 - Present

- Built a geospatial ML pipeline scoring wildfire dryness risk from NDVI/NDMI, topography, and real-time weather.
- Trained gradient-boosted trees with cross-validation; surfaced SHAP explanations to make driver features transparent.
- Deployed tiled risk maps and alert thresholds to flag prescribed-burn zones, letting responders focus on the driest 10-20% of areas; dashboards with auto-refresh cut manual triage time by ~40%.
- Calibrated thresholds against historical burns and on-the-ground moisture checks to reduce false positives in low-risk zones.

PROJECTS

CityX Subway Display System 🚇

Java, Swing, MySQL, REST APIs, Multithreading

Calgary, AB

June 2025 - July 2025

- Engineered a real-time subway screen with animated tracking and API feeds, scoring scalable commuter updates.
- Structured a MySQL ad scheduler with indexed rules, cutting query latency by 35% and eliminating duplicate lookups.
- Integrated weather and news APIs with caching layers and retry logic, maintaining 90% reliability during outages.
- Designed a modular Swing UI with multithreaded pipelines, enabling seamless ad-to-map transitions.

Flight Operations Manager 🛩️

C++, Qt, SQLite/CSV, Data Visualization

Calgary, AB

February 2025 - March 2025

- Developed a Qt desktop tool to manage flights, passengers, and schedules, handling 500+ operational records.
- Created an interactive seat map with validation layers and conflict checks, reducing booking errors by 40%.
- Optimized data storage and SQL queries to scale for larger datasets, achieving ~35% faster execution times.
- Enhanced system stability by debugging segmentation faults and applying error-handling mechanisms, sustaining ~100% uptime.

TechVista Inc 🏢

React, Flask, SQLite, REST APIs, Responsive UI

Calgary, AB

October 2024 - December 2024

- Implemented a full-stack e-commerce platform with secure login, catalog, and cart modules, improving reliability.
- Built Flask REST APIs with schema validation and parallelized endpoints, lowering response latency by 30%.
- Crafted reusable React components with hooks and context, boosting UI responsiveness while minimizing re-renders.
- Strengthened security with session-based authentication and guarded routes, preventing unauthorized access attempts.

Art Museum Data Manager 🎨

Python, MySQL, SQL Scripting

Calgary, AB

February 2024 - April 2024

- Architected a normalized MySQL schema with constraints, streamlining records across four data domains.
- Programmed a Python CLI to provision accounts and manage stable role-based access for admins and employees.
- Automated SQL scripts for seeding, reporting, and role setup, saving 5+ hours of manual work weekly.
- Enforced business rules with SQL triggers, preventing invalid updates and overdue loan extensions.